

The blackout-box mystery

Teacher guide and worked answers

Explain why an object needs light reaching it and then the eye to be seen.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsb/inputs/Build SOW 2026-2027.xlsx](#) | BUILD Weekly - Spring | C33

Recognise that we need light to see and that dark is the absence of light.

Direct match to the named outcome. The lesson is a 40-minute exemplar, with optional greater depth; no accreditation completion is claimed.

Prior learning

Everyday objects can be described by what we observe.

Use a labelled diagram or evidence card to explain a choice.

Success evidence

Identify that light is needed to see.

Explain that light enters the eye.

Distinguish dim light from complete darkness.

Equipment

Low-power battery LED torch; white card; familiar small object; optional pre-made viewing box; printed evidence.

Preparation

Prepare a box with separate viewing and light openings, both adult-cut. Keep the room comfortably lit.

Test visibility before the lesson. The box may leak light: treat this as a limitation, not pupil error.

Practical care

No dark-room requirement or blindfolds. Keep exits and walkways lit.

Use a cool, low-power LED; never aim it into eyes. No lasers, hot lamps or direct Sun viewing.

Adult prepares any holes. Offer the paper route to anyone uncomfortable with a viewing box.

Ready to teach alternative

Use the four constructed box cases; all tasks can be completed without looking into a box.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	An object is inside. Why can't we see it?
2	Retrieval	2-5	What do we already know?

Slide	Stage	Minutes	Focus
3	I do	5-9	I follow the light
4	I do	9-12	I remove the source
5	I do	Within stage	I block one part of the path
6	We do	12-16	The case conference
7	We do	16-20	Repair the explanation
8	Hinge	20-23	What makes seeing possible?
9	You do	23-25	Choose your investigation route
10	You do	Within stage	Use the evidence
11	You do	25-35	Make your own science record
12	You do	Within stage	Check and improve
13	Review	35-38	Lundy loop: improve our account
14	Exit	38-40	Three quick checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening An object is inside. Why can't we see it?

Show the optional box or describe its two openings. Do not reveal the object by touch. State the objective and offer paper-only observation.

2 Retrieval What do we already know?

Accept lit lamp, Sun or other valid source. Plain card reflects rather than produces visible light. Eyes enable sight; other senses can identify an object without seeing it.

3 I do I follow the light

Think aloud: "I start at the source, not the eye. I trace light to the object, then into the eye. Now the object can be seen." Arrows are selected light paths, not all light. Never demonstrate by shining into anyone's eye.

4 I do I remove the source

Think aloud: "I keep the object in place. In this ideal model, no light comes from anywhere. I remove the light arrows. I cannot infer the object has gone merely because it cannot be seen." A real box can leak dim light.

5 I do I block one part of the path

Shares I do time. Think aloud: "The card blocks our source. If room light entered elsewhere, we might still see the object. My model therefore says no other light."

6 We do The case conference

Partners take predictor and evidence-reader roles. A and B use the same white object. The source changes; the object remains. Invite an individual choice before the shared answer.

Reveal: B adds a light path from source to object to eye.

7 We do Repair the explanation

Read cards C and D from the shared page. Ask pairs to replace “eyes make more light” with an accurate explanation. A faint view means some light reaches the eye. Complete darkness stays without visible detail.

Reveal: C has a little light. D has none; waiting alone does not add light.

8 Hinge What makes seeing possible?

If pupils choose eyes sending light, trace the final arrow into the eye together. If they choose waiting, compare dim and no-light cases.

A. Our eyes send light onto the object. - For seeing, light enters our eyes; they do not illuminate the object.

B. Light from the object enters our eyes. - The object can give out light or reflect light from a source.

C. Waiting makes light appear in total darkness. - Adaptation helps with dim light, but it cannot create light.

9 You do Choose your investigation route

Give one route only, selected with the pupil. The shared evidence page supports every route. Optional observation: compare a covered light opening with the LED lighting the object. Never claim a practical box is perfectly dark without evidence. Pupils can use only the cases.

10 You do Use the evidence

Shares the 12-minute independent stage. Optional observation: compare a covered light opening with the LED lighting the object. Never claim a practical box is perfectly dark without evidence. Pupils can use only the cases.

11 You do Make your own science record

Allow ten minutes of independent response after the two-minute setup. Optional observation: compare a covered light opening with the LED lighting the object. Never claim a practical box is perfectly dark without evidence. Pupils can use only the cases.

12 You do Check and improve

Shares the independent stage. Ask the pupil to point to the clue supporting their answer; accept a spoken correction or changed choice.

13 Review Lundy loop: improve our account

Give a private or paired option. Adopt one scientifically sound pupil revision in the class explanation and name the change. No public ranking.

Reveal: Light reaches the object, and some reflected light enters the eye.

14 Exit Three quick checks

Collect brief individual answers. Secure core requires the main idea and a relevant reason. Revisit the relevant worked diagram next lesson if the explanation reverses cause and effect.

Reveal: See the teacher answer guide for the three checks.

Answers and assessment

Retrieval 1-3

1 Lit lamp, Sun or another valid source. 2 No, ordinary card reflects light. 3 Sight.

We do W1-W2

W1 The torch provides light; the object stays. W2 C has a small light leak; D excludes all light. Seeing C does not show that eyes generate light.

Hinge

The second option is correct. Light enters eyes. Waiting helps adaptation only when some light is present.

S1-S5

S1 B. S2 light. S3 eye. S4 dim. S5 No light has been added; eyes cannot see the object without light.

T1-T3

T1 A: no light path; B: torch to object to eye; C: some room light enters and reflects; D: no light, even after waiting. T2 arrows source → object → eye. T3 the object remains present but is not visible.

X1-X4

X1 Seeing the non-glowing object means some light reaches the eye. X2 a gap at the lid, light opening or viewing opening. X3 no visible view of the object. X4 touch uses a different sense, so it can support the object's presence without showing a light path.

Exit E1-E3

E1 light. E2 enters. E3 no. Accept a diagram or spoken explanation.

R1-R2

Responses vary. Credit a science-related revision or relevant next question. A pass or private response is acceptable; preferences have no fixed right answer.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 7; We do 8; hinge 3; You do 12; review 3; exit 2. Zero-minute slides share the stage time.

BUILD follows the user's Year 3 science anchor with age-respectful materials for secondary pupils. Routes are adjustable access choices, not fixed groups or diagnoses.

Every core task can run from the printed evidence. Do not present constructed teaching evidence as observations pupils made. A pupil's spoken reasoning carries the same scientific value as writing.

Secure core: light is needed and enters the eye. The distinction between dim and complete darkness is an explanatory extension; do not require advanced eye adaptation terminology.

HTML We do: pupils choose a prediction, test it against the user-controlled SVG model or supplied evidence, then explain or revise. The shared paper cases and W1/W2 prompts provide the equivalent classroom discussion without a device. Any animation is a teaching model, not filmed experimental evidence.

Access and responsive teaching

Supported

Choose one route: short choices, word bank and pointing or adult-scribed explanation; keep the same science goal.

Standard

Use the evidence provided to record a decision and explain the science in short sentences.

Stretch

Apply the same core idea to a changed setup, identify a limitation and justify a better decision.

Regulation

Preview the sequence and equipment. Offer seated observation, a quiet paper route, partner rehearsal and a pass. Never require public performance or a darkened room.

Misconceptions to check

Eyes send out light to find objects.

Light enters the eye from a source or a reflecting object.

Check: Ask which way the arrow points at the eye.

Waiting lets you see when there is no light at all.

Eyes adapt to dim light; they cannot see an object in complete darkness.

Check: Compare a dim room with the ideal no-light case.

Word	Meaning
light source	Something that gives out its own light.
reflect	To send light back from a surface.
darkness	An absence of light; a dim place still has some light.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 light outcomes anchor the BUILD content; ray explanations are carefully modelled as an accessible extension.